

2014 F=ma Exam: Problem 16

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We study the two limiting cases of a perfectly inelastic collision and an elastic collision which will provide us with the bounds that apply to any generic collision.



For a perfectly inelastic collision, the masses stick together so

$$m_1 v_0 = (m_1 + m_2) v_1$$
$$r_1 = \frac{v_1}{v_0} = \frac{m_1}{m_1 + m_2} = \frac{1}{1 + \alpha}$$

For an elastic collision, recall

$$v_1 = \frac{m_1 - m_2}{m_1 + m_2} v_0$$
$$r_1 = \frac{v_1}{v_0} = \frac{m_1 - m_2}{m_1 + m_2} = \frac{1 - \alpha}{1 + \alpha}$$

Thus,

$$\frac{1 - \alpha}{1 + \alpha} \leq r_1 \leq \frac{1}{1 + \alpha}$$

so the answer is B.